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Exp-12: Implement Iris Flower Classification using KNN

AIM

To implement Iris flower classification using the K-Nearest Neighbors (KNN) algorithm and classify iris flowers into different species based on their sepal and petal measurements.

ALGORITHM

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Separate the dataset into features (X) and target labels (Y).
4. Split the dataset into training and testing data.
5. Normalize the feature values.
6. Create a KNN classifier with a suitable value of K, such as K = 5.
7. Train the KNN model using the training data.
8. Predict the classes of the test data.
9. Calculate the classification accuracy.
10. Display the predicted class and compare it with the actual class.
11. Use the trained model to classify a new iris flower.

PROGRAM

```
# Import required libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score, confusion_matrix

# Load the Iris dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split the dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Create the KNN model

knn = KNeighborsClassifier(n_neighbors=3)

# Train the model

knn.fit(X_train, y_train)
```

```

# Predict the test data
y_pred = knn.predict(X_test)

# Display results
print("Actual Labels:")
print(y_test)
print("\nPredicted Labels:")
print(y_pred)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy: {:.2f}%".format(accuracy * 100))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

```

Output:

```

... Actual Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Predicted Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Accuracy: 100.00%

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

```

SAMPLE INPUT

For a new iris flower:

Feature	Value
Sepal Length	5.1 cm
Sepal Width	3.5 cm
Petal Length	1.4 cm
Petal Width	0.2 cm
K value	5

SAMPLE OUTPUT

K = 5

Predicted Class: Iris-setosa

Actual Class: Iris-setosa

Accuracy: 96.67%

RESULT

The **Iris flower classification** was successfully implemented using the **KNN algorithm**, and the model correctly classified the iris flower species with high accuracy.

